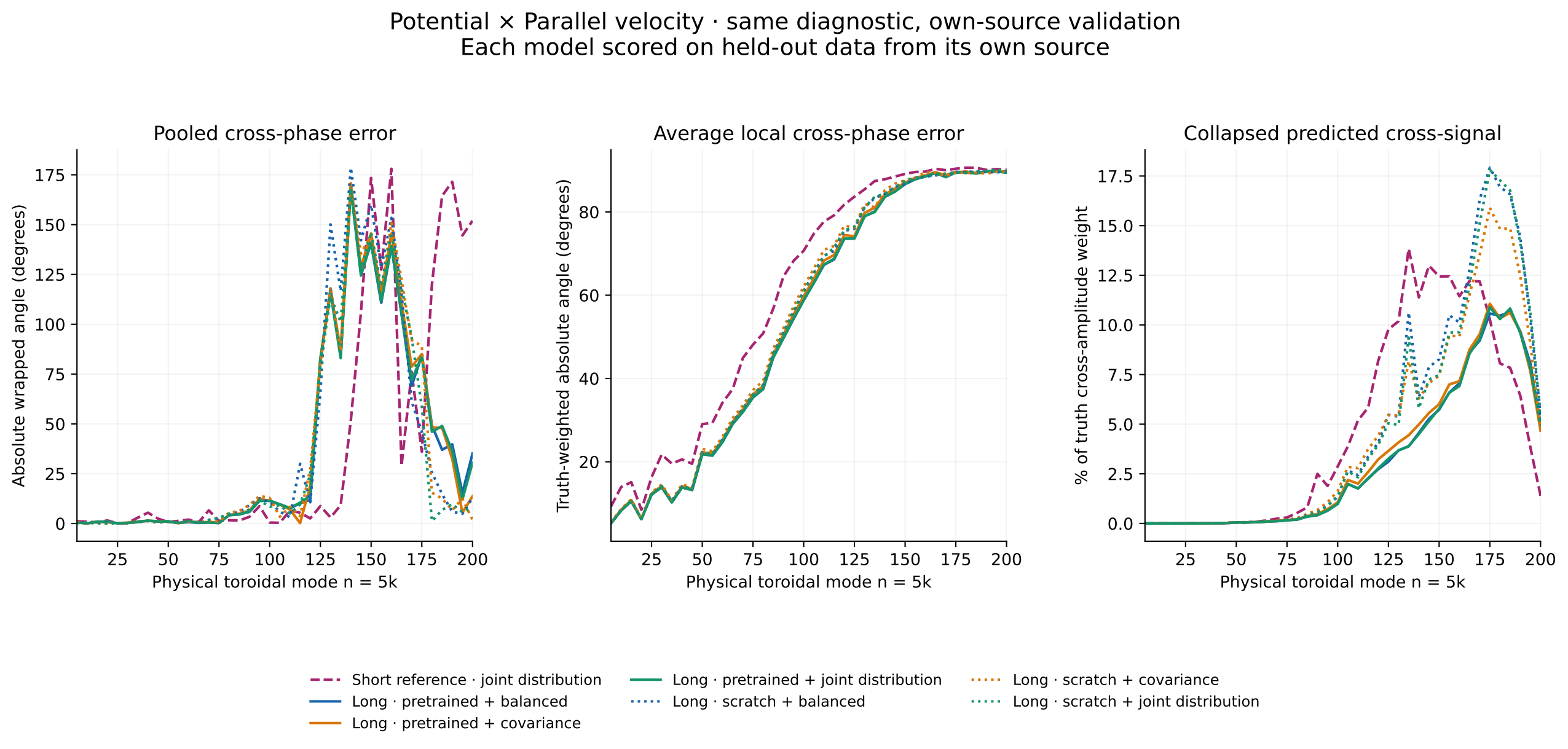
Potential × Parallel velocity · same diagnostic, own-source validation
Each model scored on held-out data from its own source
Pooled cross-phase error
Absolute wrapped angle (degrees)
Physical toroidal mode n = 5k
Average local cross-phase error
Truth-weighted absolute angle (degrees)
Physical toroidal mode n = 5k
Collapsed predicted cross-signal
% of truth cross-amplitude weight
Physical toroidal mode n = 5k
Short reference · joint distribution
Long · pretrained + balanced
Long · pretrained + covariance
Long · pretrained + joint distribution
Long · scratch + balanced
Long · scratch + covariance
Long · scratch + joint distribution